

SYMARMA: a new dynamic model for temporal data on conditional symmetric distribution

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Abstract Gaussian models of time series, ARMA, have been widely used in the literature. Benjamin et al. (J Am Stat Assoc 98:214–223, 2003) extended these models to the exponential family distributions. Also in that direction, Rocha and Cribari-Neto (Test 18:529–545, 2009) proposed a time series model for the class of beta distributions. In this paper, we develop an autoregressive and moving average symmetric model, named SYMARMA, which is a dynamic model for random variables belonging to the class of symmetric distributions including also a set of regressors. We discuss methods for parameter estimation, hypothesis testing and forecasting. In particular, we provide closed-form expressions for the score function and Fisher information matrix. Robust study is presented based on influence function. We conduct simulation studies to evaluate the consistency and asymptotic normality of the conditional maximum likelihood estimator for the model parameters. An application with real data is presented and discussed.

Keywords Conditional maximum likelihood · Outlier · Symmetric distributions · Time series

1 Introduction

Nature events, economic index, consequences of the development of a society, among other facts, produce information that require a sophisticated analysis of data to under-

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stand the dynamics of such phenomena and thus make inferences that enable a decision making with more accurately. In almost all areas of knowledge, there are phenomena of interest that develop and change over time. Data collected at different time instants, in general, leads to new problems that have no standard methodology to be considered. The presence of outliers in time series, for example, may compromise the inferential procedure of statistical methods more conventional that assume a normal distribution for the errors, such as the usual stationary linear models (ARMA), and may provide enough biased estimators of the parameters or even a misspecification of the model.

Some works have been developed on time series models using alternative ways to normality assumption. The pioneer studies in non-Gaussian time series was developed by Heyde and Feigin (1975) who developed Poisson conditional autoregressive range. Cox (1981) proposed a methodology based on two approaches: models conditioned on past observations and latent processes. Zeger (1988) extended this model including the past and present of covariates. Zeger and Qaqish (1988) developed Markov chain for time series. Li (1994) included moving averages models component in this models. Peña (1990) developed methods to identify influential observations in univariate ARIMA models. More recently, Benjamin et al. (2003) extended ARMA models to data whose conditional distribution given the past of the process belongs to the exponential family, named GARMA model. In the same approach, Rocha and Cribari-Neto (2009) developed dynamic models in the beta family distribution (β ARMA). Cao et al. (2010) discuss heteroscedasticity and/or autocorrelation tests in nonlinear models with AR(1) in symmetric class of distributions. Creal et al. (2013) proposed a class of observation-driven time series models referred to as generalized autoregressive score (GAS) models.

The main goal of this work is to propose a class of models under symmetrical distribution to time series data, named SYMARMA model. This class distributions contains with lighter and heavier tails than the normal one. Distributions with heavier tails tend to better accommodate atypical observations. The paper is organized into 7 sections. In Sect. 2, we introduce the SYMARMA model and present the stationarity conditions for marginal mean and variance. In Sect. 3, we develop closed expressions for the score function and Fisher information, estimation method and hypotheses test. Monte Carlo simulation studies to evaluate the performance of the estimators of the parameter SYMARMA model and robustness aspects based on influence function are presented in Sect. 4. In addition, we also evaluate the performance of the proposed model based on mean square error in presence of additive and innovation outliers. Forecasting techniques are discussed in Sect. 5. We carry out an application with real data to illustrate the proposed methodology in Sect. 6. Finally, in last section, the concluding remarks are presented.

2 Symmetric autoregressive moving average model

Let Y_t , for $t = 1, \dots, n$, be a random variable and assume that the conditional distribution of each Y_t , given the set of past information $\mathcal{F}_{t-1} =$

$(Y_{t-1}, \dots, Y_1, \mu_{t-1}, \dots, \mu_1)$, follows a continuous symmetric distribution, denoted by $Y_t | \mathcal{F}_{t-1} \sim S(\mu_t, \phi, g)$ and its density takes the form

$$f(y_t | \mathcal{F}_{t-1}, \mu_t, \varphi) = \frac{1}{\sqrt{\varphi}} g \left\{ \frac{(y_t - \mu_t)^2}{\varphi} \right\}, \quad y_t \in \mathbb{R}, \quad \mu_t \in \mathbb{R} \quad \text{and} \quad \varphi > 0, \quad (1)$$

where μ_t is the location parameter, φ is the scale parameter and $g(\cdot)$, is known as the generating function of densities, with $g(u) > 0$, for $u > 0$, such that $\int_0^\infty u^{-\frac{1}{2}} g(u) du = 1$. When they exist, the mean and variance conditional of Y_t are μ_t and $\xi\varphi$, respectively, where $\xi > 0$ is a constant that depends on characteristic function of the each symmetric distribution. For example, we have $\xi = 1$ for normal distribution, $\xi = \nu/(\nu - 2)$ for Student t distribution with ν degrees of freedom ($\nu > 2$) and $\xi = 0.7957$ for logistic-I distribution.

The SYMARMA model is given by (1) and

$$\mu_t = \mathbf{x}_t^\top \boldsymbol{\beta} + \tau_t$$

with $\mathbf{x}_t^\top = (1, x_{t1}, x_{t2}, \dots, x_{tk})$ is the k explanatory variables values vector, $k < n$, and $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \dots, \beta_k)^\top$ is the unknown parameter vector and τ_t is the dynamic component with autoregressive moving average terms included additively. In the SYMARMA model, τ_t is defined by

$$\tau_t = \sum_{i=1}^p \phi_i \{y_{t-i} - \mathbf{x}_{t-i}^\top \boldsymbol{\beta}\} + \sum_{j=1}^q \theta_j r_{t-j},$$

where ϕ_i 's, θ_j 's, p and $q \in \mathbb{N}$ are autoregressive and moving average parameters and their orders, respectively. Here, r_t is uncorrelated random errors measurable with respect to $\mathcal{F}_t$. For instance, we can consider $r_t = y_t - \mu_t$ (errors in the original scale of measurement), Pearson residuals, among others. Therefore, we have μ_t is given by

$$\mu_t = \mathbf{x}_t^\top \boldsymbol{\beta} + \sum_{i=1}^p \phi_i \{y_{t-i} - \mathbf{x}_{t-i}^\top \boldsymbol{\beta}\} + \sum_{j=1}^q \theta_j r_{t-j}. \quad (2)$$

The SYMARMA(p, q) model is defined by the random component with density given in (1) and the dynamic component given in (2). Standard ARMA(p, q) and standard ARMAX(p, q) models are particular cases from the SYMARMA model with p, q parameters.

2.1 Stationarity conditions

Theorem 1 *The marginal mean of Y_t in the SYMARMA model is given by*

$$E(Y_t) = \mathbf{x}_t^\top \boldsymbol{\beta},$$

provided that $\Phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p$ is invertible, where B^k is the lag operator (i.e., $B^k y_t = y_{t-k}$).

Theorem 2 *The marginal variance of Y_t in the SYMARMA model is given by*

$$\text{Var}(Y_t) = \xi \varphi \sum_{i=0}^{\infty} \psi_i^2,$$

where $\Psi(B) = \sum_{i=0}^{\infty} \psi_i B^i = \Theta(B)\Phi(B)^{-1}$, $\psi_0 = 1$ and assuming that $\Phi(B)$ is invertible.

Theorem 3 *The covariance and correlation of Y_t and Y_k in the SYMARMA model are given by*

$$\text{Cov}(Y_t, Y_{t-k}) = \xi \varphi \sum_{i=0}^{\infty} \psi_i \psi_{i+k}, \quad \text{and} \quad \text{Corr}(Y_t, Y_{t-k}) = \frac{\sum_{i=0}^{\infty} \psi_i \psi_{i+k}}{\sum_{i=0}^{\infty} \psi_i^2}$$

According to Theorem 1, we have that the marginal mean of Y_t in a SYMARMA model is stationary since $\Phi(B)$ is invertible and $\mathbf{x}_t^\top \boldsymbol{\beta} = c$, for all t with c being a constant. Whereas, by Theorem 2, the condition of stationarity for marginal variance of Y_t in a SYMARMA model is also when $\Phi(B)$ is invertible. Proofs of Theorems 1 – 3 are presented in Appendix.

3 Estimation and testing

The estimation of the parameters that index the SYMARMA model can be obtained by maximum conditional likelihood method on the first m observations, where $m = \max\{p, q\}$. The conditional likelihood function is a function of the parameters β 's, ϕ 's, θ 's and φ , given by

$$L(\boldsymbol{\delta}, \varphi) = \prod_{t=m+1}^n f(y_t | \mathcal{F}_{t-1}, \mu_t, \varphi),$$

where $\boldsymbol{\delta} = (\boldsymbol{\beta}^\top, \boldsymbol{\phi}^\top, \boldsymbol{\theta}^\top)^\top$ and, the conditional log-likelihood function is expressed by

$$\ell(\boldsymbol{\delta}, \varphi) = \sum_{t=m+1}^n \log f(y_t | \mathcal{F}_{t-1}, \mu_t, \varphi) = \sum_{t=m+1}^n -\frac{1}{2} \log(\varphi) + \log \{g(u_t)\}, \quad (3)$$

where $u_t = \frac{(y_t - \mu_t)^2}{\varphi}$ and μ_t is defined in (2).

The conditional log-likelihood function, ℓ , is assumed to be regular (Cox and Hinkley 1974, Cap. 9) with respect to δ and φ . Furthermore, we require to differentiate ℓ with respect to the unknown parameters in order to obtain the score function and the Fisher information matrix. Symmetric distributions that do not satisfy the usual regularity conditions are not considered in this paper, for example, the Kotz, generalized Kotz and double exponential distributions.

In the construction of the conditional log-likelihood function, the first q errors are assumed to be equal zero since that conditional to $\mathcal{F}_m$, the first m errors are zero (or approximately zero). Expressions for the score vector ($\mathbf{U}$) and conditional Fisher information matrix ($\mathbf{K}$) are obtained, respectively, by partial derivatives of first and second order of the logarithm of the conditional likelihood function given in (3). Assuming that the function $g(\cdot)$ is continuous and differentiable, score functions for parameter vector δ and φ are given, respectively, by

$$\begin{aligned} \mathbf{U}_\delta(\delta, \varphi) &= \varphi^{-1} \mathbf{O}^\top \mathbf{D}(\mathbf{y} - \boldsymbol{\mu}) \quad \text{and} \\ \mathbf{U}_\varphi(\delta, \varphi) &= \frac{1}{2\varphi} [-(n-m) + \varphi^{-1} (\mathbf{y} - \boldsymbol{\mu})^\top \mathbf{D}(\mathbf{y} - \boldsymbol{\mu})], \end{aligned}$$

where $\mathbf{y} = (y_{m+1}, \dots, y_n)^\top$, $\boldsymbol{\mu} = (\mu_{m+1}, \dots, \mu_n)^\top$, $\mathbf{D} = \text{diag}\{v_{m+1}, \dots, v_n\}$ with $v_t = v(u_t) = -2W_g(u_t)$ and $W_g(u_t) = \partial \log(g(u_t)) / \partial u_t$. Also $\mathbf{O} = (\mathbf{C}, \mathbf{A}, \mathbf{B})$ is a $(n-m) \times (p+q+k+1)$ matrix with

$$\begin{aligned} \mathbf{A} &= \begin{pmatrix} y_m - \mathbf{x}_m^\top \boldsymbol{\beta} & \dots & y_{m+1-p} - \mathbf{x}_{m+1-p}^\top \boldsymbol{\beta} \\ y_{m+1} - \mathbf{x}_{m+1}^\top \boldsymbol{\beta} & \dots & y_{m+2-p} - \mathbf{x}_{m+2-p}^\top \boldsymbol{\beta} \\ \vdots & \ddots & \vdots \\ y_{n-1} - \mathbf{x}_{n-1}^\top \boldsymbol{\beta} & \dots & y_{n-p} - \mathbf{x}_{n-p}^\top \boldsymbol{\beta} \end{pmatrix}_{(n-m) \times p}, \\ \mathbf{B} &= \begin{pmatrix} y_m - \mu_m & \dots & y_{m+1-q} - \mu_{m+1-q} \\ y_{m+1} - \mu_{m+1} & \dots & y_{m+2-q} - \mu_{m+2-q} \\ \vdots & \ddots & \vdots \\ y_{n-1} - \mu_{n-1} & \dots & y_{n-q} - \mu_{n-q} \end{pmatrix}_{(n-m) \times q}, \\ \mathbf{C} &= \begin{pmatrix} 1 - \sum_{i=1}^p \phi_i & \dots & x_{(m+1)k} - \sum_{i=1}^p \phi_i x_{(m+1-i)k} \\ 1 - \sum_{i=1}^p \phi_i & \dots & x_{(m+2)k} - \sum_{i=1}^p \phi_i x_{(m+2-i)k} \\ \vdots & \ddots & \vdots \\ 1 - \sum_{i=1}^p \phi_i & \dots & x_{(n)k} - \sum_{i=1}^p \phi_i x_{(n-i)k} \end{pmatrix}_{(n-m) \times (k+1)}. \end{aligned}$$

Table 1 Expressions for $W_g(u_t)$ and $W'_g(u_t)$ for some symmetric distributions

Distribution	$W_g(u_t)$	$W'_g(u_t)$
Normal	$-\frac{1}{2}$	0
Student t	$-\frac{(v+1)}{2(v+u)}$	$\frac{(v+1)}{2(v+u)^2}$
Generalized Student t	$-\frac{(r+1)}{2(s+u)}$	$\frac{(r+1)}{2(s+u)^2}$
Logistic I	$-\tanh\left(\frac{u}{2}\right)$	$-\operatorname{sech}\left(\frac{u}{2}\right)/2$
Logistic II	$-\frac{\exp(-\sqrt{u})-1}{(-2\sqrt{u})[1+\exp(-\sqrt{u})]}$	$\frac{2\exp(-\sqrt{u})\sqrt{u}+\exp(-2\sqrt{u})-1}{-4u^{3/2}[1+\exp(-\sqrt{u})]^2}$
Generalized logistic	$-\frac{\alpha m[\exp(-\alpha\sqrt{u})-1]}{(-2\sqrt{u})[1+\exp(-\alpha\sqrt{u})]}$	$-\frac{\alpha m}{4} \frac{2\alpha\exp(-\alpha\sqrt{u})\sqrt{u}+\exp(-2\alpha\sqrt{u})-1}{u^{3/2}[1+\exp(-\alpha\sqrt{u})]^2}$
Power exponential	$-\frac{1}{2(1+k)u^{k/(k+1)}}$	$\frac{k}{2(1+k)^2u^{(2k+1)/(k+1)}}$

Expressions for $W_g(u_t)$ and $W'_g(u_t) = \partial W_g(u_t)/\partial u_t$ can be found in [Cysneiros and Paula \(2005\)](#) for some symmetric distributions, (see Table 1).

The conditional maximum likelihood estimator (CMLE), $\hat{\delta}$, to vector parameter δ , is obtained by $\mathbf{U}(\delta) = 0$. Except for $\mathbf{U}(\varphi)$, the system of equations has not explicit solution and thus we must use an iterative procedure to estimate the parameters, such as Newton-Raphson, Fisher scoring, among others. The Fisher scoring method has been used by several authors to estimate the parameters of symmetric regression models (see, for example, [Galea et al. 2003](#) and [Cysneiros and Paula 2005](#)). This work also is used the Fisher scoring method for estimating of the parameters that index the SYMARMA model. The iterative procedure is given by

$$\delta^{(i+1)} = \delta^{(i)} + (\mathbf{K}^{(i)})^{-1} \mathbf{U}^{(i)}, \quad i = 1, 2, \dots$$

where $\mathbf{K}$ is the expected conditional Fisher information matrix for (δ, φ) and $\mathbf{K}^{-1}$ is the inverse matrix of $\mathbf{K}$. It may be showed that $\mathbf{K}_{\delta, \varphi} = \mathbf{0}$, i.e., δ and φ are orthogonal parameters (see, details in Appendix). The expected conditional Fisher information matrix are expressed by $\mathbf{K}_{\delta\delta} = 4d_g\varphi^{-1}\mathbf{O}^\top\mathbf{O}$ and $\mathbf{K}_{\varphi\varphi} = \frac{n-m}{4\varphi^2}(4f_g - 1)$ with $d_g = \mathbf{E}\left[W_g^2(u_t)u_t|\mathcal{F}_{t-1}\right]$ and $f_g = \mathbf{E}\left[W_g^2(u_t)u_t^2|\mathcal{F}_{t-1}\right]$. Values of d_g and f_g for some symmetric distributions are listed in [Cysneiros and Paula \(2005\)](#), (see Table 2).

The iterative process algorithm is given by:

- Let be $i = 0$ initialize values for $\delta^{(i)}$ based on the estimates obtained by fitting the Gaussian ARMA(p, q) model.
- iterative equations

$$\begin{aligned} \delta^{(i+1)} &= \left(\mathbf{O}^{\top(i)}\mathbf{O}^{(i)}\right)^{-1} \mathbf{O}^{\top(i)}z_{\delta}^{(i)} \\ \varphi^{(i+1)} &= \frac{1}{n-m} \left(\mathbf{y} - \boldsymbol{\mu}^{(i+1)}\right)^\top \mathbf{D}^{(i+1)} \left(\mathbf{y} - \boldsymbol{\mu}^{(i+1)}\right), \end{aligned}$$

Table 2 Expressions for d_g and f_g for some symmetric distributions

Distribution	d_g	f_g
Normal	$\frac{1}{4}$	$\frac{3}{4}$
Student t	$\frac{(v+1)}{4(v+3)}$	$\frac{3(v+1)}{4(v+3)}$
generalized Student t	$\frac{r(r+1)}{4s(r+3)}$	$\frac{3(r+1)}{4(r+3)}$
Logistic I	0,36931	1,00345
Logistic II	$\frac{1}{12}$	0,60749
Generalized logistic	$\frac{\alpha^2 m^2}{4(2m+1)}$	$\frac{2m(2+m^2\psi'(m))}{4(2m+1)}$
Power exponential	$\frac{\Gamma\{(3-k)/2\}}{4(2^{k-1})(1+k)^2\Gamma\{(k+1)/2\}}$	$\frac{(k+3)}{4(k+1)}$

with z_δ a vector $(n - m) \times 1$ expressed as

$$z_\delta = \mathbf{O}\delta + (4d_g)^{-1}\mathbf{D}(\mathbf{y} - \mu).$$

(c) Given $\epsilon > 0$, if $\|\delta^{(i+1)} - \delta^{(i)}\| \leq \epsilon$ then stop else $i = i + 1$ and go to (b).

3.1 Inferences in the SYMARMA model

Under the usual regularity conditions and for n sufficiently large, we have the conditional maximum likelihood estimator converges in distribution to a normal distribution, i.e.,

$$\begin{pmatrix} \hat{\delta} \\ \hat{\varphi} \end{pmatrix} \sim N_{(k+p+q+2)} \left(\begin{pmatrix} \delta \\ \varphi \end{pmatrix}, \mathbf{K}^{-1} \right), \quad (4)$$

where N_r denotes the r -dimensional normal distribution. $\hat{\delta}$ and $\hat{\varphi}$ are the conditional maximum likelihood estimators of δ and φ , respectively. $\mathbf{K}_{\delta\delta}^{-1}$ evaluated at $(\hat{\delta}, \hat{\varphi})$ and $\mathbf{K}_{\varphi\varphi}^{-1}$ evaluated at $(\hat{\delta}, \hat{\varphi})$ are consistent estimator of the asymptotic variance of $\hat{\delta}$ and of $\hat{\varphi}$, respectively. Based on the asymptotic normality of the CMLE of (δ, φ) , $(\hat{\delta}, \hat{\varphi})$, given in (4), approximate confidence intervals can be easily constructed.

Hypotheses involving parameter vector $\zeta = (\beta^\top, \phi^\top, \theta^\top, \varphi)^\top$ can be expressed in the general form

$$\mathcal{H}_0 : \mathbf{Z}\zeta = \mathbf{0} \quad \text{and} \quad \mathcal{H}_1 : \mathbf{Z}\zeta \neq \mathbf{0}, \quad (5)$$

where $\mathbf{Z}$ is a $r \times (k + p + q + 2)$ matrix of full rank r ($r < k + p + q + 2$). The null hypothesis ($\mathcal{H}_0$) can look fairly simple situations, such as testing $\mathcal{H}_0: \beta = 0$ against $\mathcal{H}_1: \beta \neq 0$. The statistic used to test the models obtained from the restrictions $\mathcal{H}_0$ and $\mathcal{H}_1$ in (5) may be the conditional likelihood ratio (CLR) given by

$$\lambda = 2 \{ \ell(\hat{\delta}) - \ell(\tilde{\delta}) \},$$

where $\ell(\cdot)$ is the logarithm of the conditional likelihood function, $\tilde{\delta}$ is the restricted CMLE of δ under $\mathcal{H}_0$ and $\hat{\delta}$ is the unrestricted CMLE of δ under $\mathcal{H}_1$. Under suitable regularity conditions, It follows under $\mathcal{H}_0$ and for large n that λ have Chi-squared distribution with r degrees of freedom.

4 Robustness, consistency and asymptotic normality

In this section we demonstrate the qualitative robustness of the estimation procedure and results of the simulation studies to evaluate the consistency and asymptotic normality of the conditional maximum likelihood estimator. In addition, we also evaluate the performance of the SYMARMA model in presence of additive and innovation outliers.

4.1 Influence function

An estimation procedure is qualitatively robust if the influence function (IF) defined by

$$\text{IF}(y, \delta, \varphi, f) = -[E[\Psi'(y, \delta, \varphi, f)]]^{-1} \Psi(y, \delta, \varphi, f) \quad (6)$$

is bounded, where Ψ is such that $\sum_{t=1}^n \Psi(y_t, \delta, \varphi, f) = 0$ produces the CMLE of (δ, φ) , Ψ' is its derivative, and f is the p.d.f. associated to model with (δ, φ) parameter.

We consider the IF defined in (6) for SYMARMA-normal and SYMARMA- t models. In order to simplify some calculations, without loss of generality, we assume $\mu_t = \mu$. The score functions associated with φ and μ for SYMARMA models are

$$\Psi(y, \varphi, f) = -\frac{1}{2\varphi} - \frac{1}{\varphi} W_g(u_t) u_t \quad \text{and} \quad \Psi(y, \mu, f) = -\frac{2}{\varphi} W_g(u_t) (y_t - \mu_t).$$

We have proved that, for the SYMARMA-normal models, the corresponding IF is proportional to its score, namely Ψ . Thus, since $\Psi(y, \varphi, f)$ and $\Psi(y, \mu, f)$ are unbounded (see Fig. 1), then their corresponding IFs are also unbounded. This aspect confirms that the procedure of parameter estimation based on the SYMARMA-normal model is non-robust providing estimates that are sensitive to atypical observations.

In the SYMARMA- t case, for ν assumed to be known, the IF for φ and μ are also proportional to its score, but here $\Psi(y, \varphi, f)$ and $\Psi(y, \mu, f)$ are bounded and consequently the corresponding IF are also bounded (see Fig. 2 with $\nu = 4$). In this way, the procedure of estimation for φ and μ based on the SYMARMA- t model is qualitatively robust.

Lucas (1997) developed an interesting study on the robust aspects of the Student t model. He showed that the protection against outliers is preserved only if the degrees of freedom extra parameter ν is fixed. Otherwise, if the degrees of freedom parameter is also estimated by maximum likelihood, the influence functions for φ and ν and the change-of-variance function of the location parameter are not bounded. In this direction, we keep all the extra parameters fixed or known, for instance, ν in Student t distribution. Similar study was performed in Paula et al. (2012).

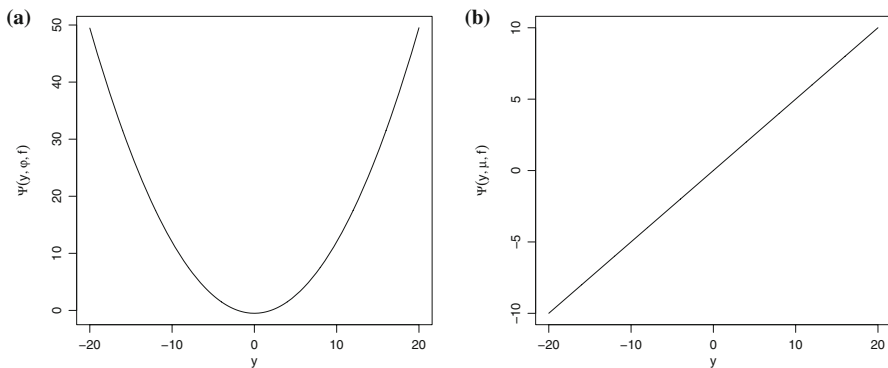


Fig. 1 Score functions for φ (a) and μ (b) of the SYMARMA-normal model

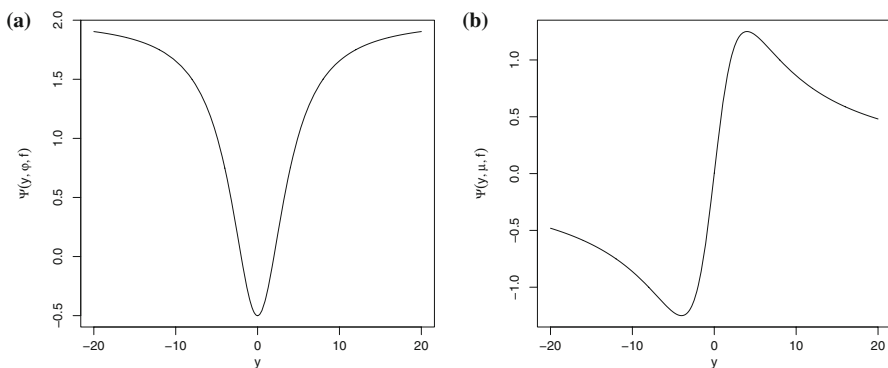


Fig. 2 Score functions for φ (a) and μ (b) of the SYMARMA- t model

4.2 Consistency and asymptotic normality

In order to evaluate the consistency and asymptotic normality of the estimators of the parameters for the SYMARMA model, we conduct a simulation studies in the SYMARMA model with intercept and one autoregressive parameter, given by

$$y_t = \beta_0 + \phi(y_{t-1} - \beta_0) + r_t, \quad t = 2, \dots, n, \quad (7)$$

where $\beta_0 = 20$, $\phi = 0.6$ and $\varphi = 1$. We use the R programming (see, www.R-project.org and R Core Team 2012) to implement all functions developed in this paper. Codes in R can be obtained from the authors upon request. We generate 10,000 Monte Carlo replicates of the model (7) under the normal, Student t and logistic I distribution for $y_t | \mathcal{F}_{t-1}$. For each scenery and combination of sample size $n = 50, 100$ and 200 we evaluate bias and mean square error (MSE) and we display normal quantile-quantile (QQ) plot for the empirical distribution of each parameter model.

From Table 3, the biases are close to zero and MSEs decrease when the sample size n increases. We emphasize that the MSEs for the $\hat{\varphi}$ in SYMARMA model under

Table 3 Bias and MSE for the CMLEs of the parameters model (7) under the normal, Student t and logistic I distribution for $y_t|\mathcal{F}_{t-1}$

Parameter	Bias			MSE		
	$n = 50$	$n = 100$	$n = 200$	$n = 50$	$n = 100$	$n = 200$
Normal						
β_0	0.0062	0.0044	0.0044	0.1273	0.0631	0.0319
ϕ	0.0574	0.0278	0.0138	0.0184	0.0075	0.0035
φ	0.0429	0.0214	0.0116	0.0403	0.0202	0.0100
Student t_4						
β_0	0.0041	0.0024	0.0001	0.1882	0.0918	0.0451
ϕ	0.0441	0.0211	0.0097	0.0139	0.0057	0.0025
φ	0.0277	0.0145	0.0046	0.0725	0.0352	0.0170
Logistic I						
β_0	0.0060	0.0045	0.0026	0.0619	0.0307	0.0152
ϕ	0.0644	0.0330	0.0169	0.0255	0.0114	0.0055
φ	0.6070	0.5964	0.5920	0.3805	0.3617	0.3534

conditional logistic I distribution is greater than in SYMARMA- t_4 and SYMARMA-normal models.

Figures 3 and 4 display the normal QQ-plot for the empirical distribution of each parameter. We can note that normal distribution approximation for $\hat{\beta}_0$, $\hat{\phi}$ and $\hat{\varphi}$ distribution are quite satisfactory. However, in SYMARMA-logistic I model we note observed that the normal distribution approximation for $\hat{\varphi}$ distribution requires larger samples.

4.3 Impact of the presence of outliers in SYMARMA models

We conduct a simulation studies in the SYMARMA(1,1) model given by

$$y_t = \phi y_{t-1} + \theta r_{t-1} + r_t, \quad t = 2, \dots, 50, \quad (8)$$

where $\phi = \{0.2, 0.4, 0.6\}$, $\theta = \{0.2, 0.3\}$ and $\varphi = 1$. We generate 10,000 Monte Carlo replicates of the model (8) under the normal and Student t_4 distribution for $y_t|\mathcal{F}_{t-1}$. For each simulated time series were introduced the additive and innovation outliers in components y_{20} and y_{40} . In both cases, the intensities of outliers were $5S_y$ and $8S_y$, where S_y denotes the standard deviation of the generated sample. The root mean square error (RMSE) is used to compare the fitted models.

Table 4 presents the proportion of cases for which the fitted SYMARMA- t_4 model obtained smaller RMSE compared to the RMSE obtained by fitted SYMARMA-normal from 10,000 simulated SYMARMA models in presence of additive and innovation outliers. In Table 5 presents the average relative gain obtained in cases where the SYMARMA- t_4 model obtained smaller RMSE. The results indicate that

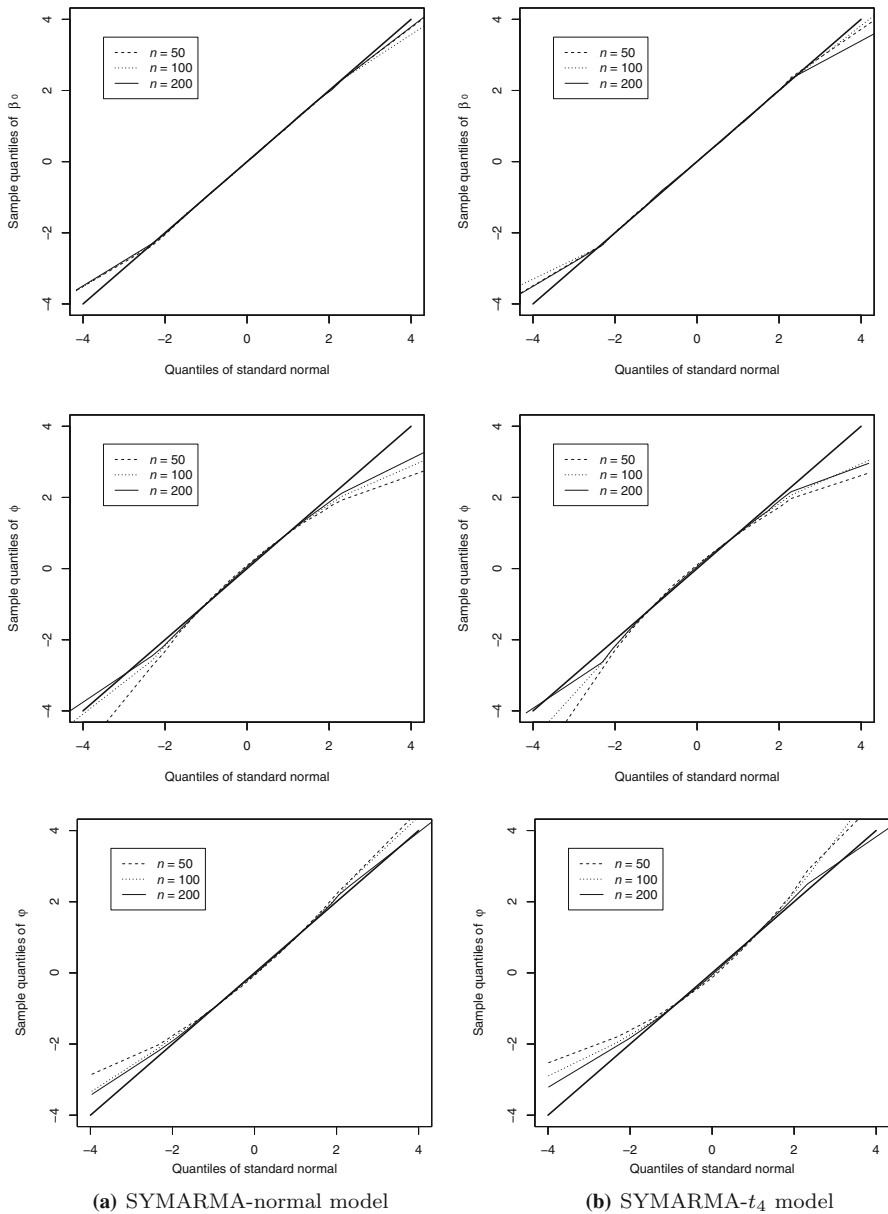


Fig. 3 Normal QQ-plots of CMLEs of the parameters β_0 (first panel), ϕ (second panel) and φ (third panel) for the SYMARMA-normal (a) and SYMARMA- t_4 (b) models (7)

the presence of atypical observations can greatly affect the estimates of model parameters, depending on sample size, the type of outlier presents, the amount and intensity of the impact(s) of outlier(s). In addition, the RMSE obtained by the SYMARMA- t_4 model in the presence of more than one outliers are smaller than the RMSE obtained

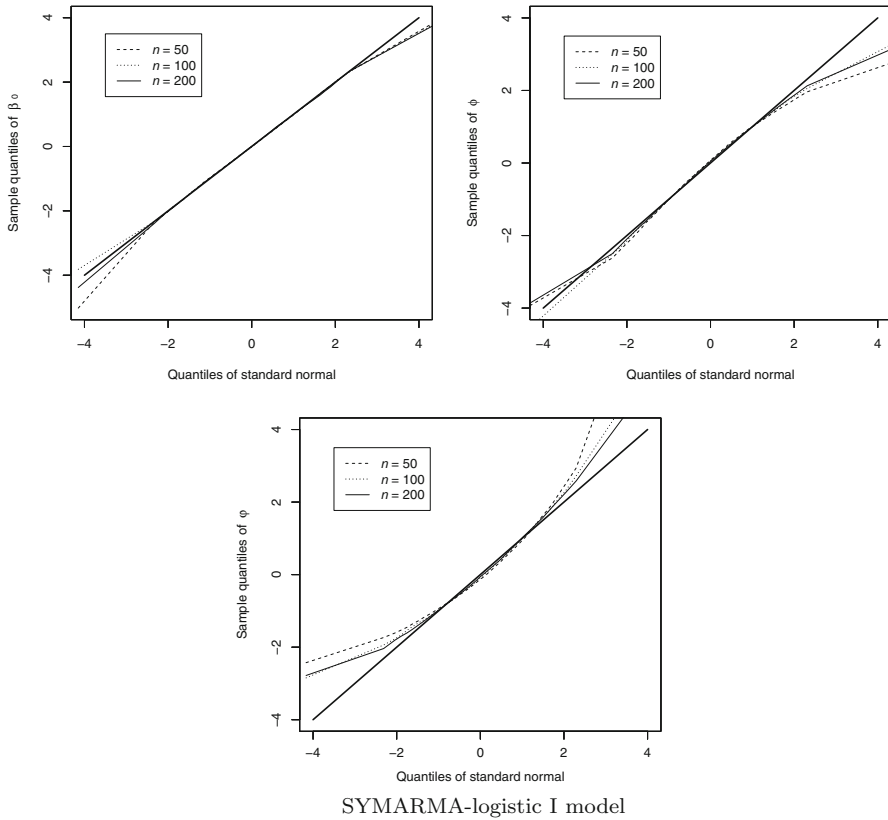


Fig. 4 Normal QQ-plots of CMLEs of the parameters β_0 , ϕ and φ for the SYMARMA-logistic I model (7)

by the SYMARMA-normal, in general. The average relative gain obtained, in many cases, is greater than 35 %. Overall, for all scenarios considered in our simulations with samples in presence of innovative outlier(s), the SYMARMA- t_4 model has a better performance, in order to less root mean square error, in more than 60 % of cases.

5 Forecasts

In order to produce forecasts, the first of all we plug in the CMLEs of δ and φ in μ_t , $t = m + 1, \dots, n$, say $\hat{\mu}_t$

$$\hat{\mu}_t = \mathbf{x}_t^\top \hat{\boldsymbol{\beta}} + \sum_{i=1}^p \hat{\phi}_i \left\{ y_{t-i} - \mathbf{x}_{t-i}^\top \hat{\boldsymbol{\beta}} \right\} + \sum_{j=1}^q \hat{\theta}_j \hat{r}_{t-j}.$$

Consequently with $\hat{\mu}_t$, we can obtain the estimates of r_t , $\hat{r}_t$, for $t = m + 1, \dots, n$. For example, for $r_t = y_t - \mu_t$, i.e., measurement errors in the original scale, we have $\hat{r}_t = y_t - \hat{\mu}_t$. For $N > n$, the forecast of the error r_N equals zero. Thus, to predict

Table 4 Percentage of cases in which the RMSE of the fitted SYMARMA- t_4 model is smaller than RMSE fitted SYMARMA-normal in presence of outlier(s) of the additive an innovation type ($n = 50$)

	θ	Position of outlier in sample	ϕ and intensities of outlier(s) (ωS_y)					
			$\phi = 0.2$		$\phi = 0.4$		$\phi = 0.6$	
			$\omega = 5$	$\omega = 8$	$\omega = 5$	$\omega = 8$	$\omega = 5$	$\omega = 8$
additive	0.2	$t = 20$	59.3	61.2	48.5	49.9	39.8	36.7
		$t = 40$	57.6	60.3	50.5	50.0	36.0	34.1
		$t = 20, 40$	65.6	70.6	61.1	70.3	51.3	67.8
	0.3	$t = 20$	55.8	57.4	43.3	43.1	36.4	34.4
		$t = 40$	53.7	55.5	43.4	42.7	33.4	31.8
		$t = 20, 40$	63.5	70.3	59.7	71.9	48.9	67.7
innovation	0.2	$t = 20$	63.6	66.2	65.5	67.9	66.8	67.4
		$t = 40$	64.7	68.5	64.5	66.9	65.7	66.1
		$t = 20, 40$	65.7	70.3	69.4	72.3	69.5	72.8
	0.3	$t = 20$	63.9	67.1	65.6	68.4	65.6	68.3
		$t = 40$	60.8	65.5	64.2	66.5	64.5	68.4
		$t = 20, 40$	65.5	68.6	69.3	73.0	70.7	74.0

Table 5 Average relative gain obtained in cases in which the RMSE of the fitted SYMARMA- t_4 model is smaller than RMSE fitted SYMARMA-normal in presence of outlier(s) of the additive an innovation type ($n = 50$)

	θ	Position of outlier in sample	ϕ and intensities of outlier(s) (ωS_y)					
			$\phi = 0.2$		$\phi = 0.4$		$\phi = 0.6$	
			$\omega = 5$	$\omega = 8$	$\omega = 5$	$\omega = 8$	$\omega = 5$	$\omega = 8$
additive	0.2	$t = 20$	35.5	66.0	29.6	58.4	29.3	67.9
		$t = 40$	33.2	57.1	30.2	56.6	29.7	59.5
		$t = 20, 40$	43.7	55.5	47.5	72.3	45.1	63.2
	0.3	$t = 20$	39.5	64.4	27.6	65.0	30.7	60.8
		$t = 40$	31.0	65.2	38.9	56.4	30.8	75.1
		$t = 20, 40$	49.6	66.4	41.8	65.9	39.6	72.1
innovation	0.2	$t = 20$	37.4	56.7	28.7	48.8	25.4	35.4
		$t = 40$	40.6	68.7	26.7	44.5	24.1	38.2
		$t = 20, 40$	54.5	50.4	32.4	45.0	36.1	35.6
	0.3	$t = 20$	36.5	57.1	35.0	41.5	24.1	37.8
		$t = 40$	32.3	63.9	27.0	43.1	21.6	38.8
		$t = 20, 40$	46.2	59.5	34.0	48.9	31.4	45.0

the mean value of the process at $N > n$, one should use the CMLEs $\hat{\delta}$ and $\hat{\phi}$; the estimates of $\mu_t, t = m + 1, \dots, n$; the estimates of $r_t, t = m + 1, \dots, n$; replace r_t by zero if $N > n$ (these suffice to obtain $\hat{\mu}_{t+1}$, and one can then proceed analogously to obtain $\hat{\mu}_{t+2}$, and so on); and replace y_t by $\hat{\mu}_t$ if $n < t < N$. Thus, the mean response estimate at $n + 1$ takes form

$$\hat{y}_{n+1} = \mathbf{x}_{n+1}^\top \hat{\beta} + \sum_{i=1}^p \hat{\phi}_i \left\{ y_{n+1-i} - \mathbf{x}_{n+1-i}^\top \hat{\beta} \right\} + \sum_{j=1}^q \hat{\theta}_j \hat{r}_{n+1-j}.$$

At time $n + 2$, we obtain

$$\hat{y}_{n+2} = \mathbf{x}_{n+2}^\top \hat{\boldsymbol{\beta}} + \sum_{i=1}^p \hat{\phi}_i \left\{ y_{n+2-i} - \mathbf{x}_{n+2-i}^\top \hat{\boldsymbol{\beta}} \right\} + \sum_{j=1}^q \hat{\theta}_j \hat{r}_{n+2-j},$$

and so on.

Bootstrap method (Efron and Tibshirani 1993) can be used to obtain the empirical distribution $\hat{F}$ of the future observation ($y_{n+\ell}$). We can construct the percentile confidence interval with approximate coverage of $(1 - \alpha)$, defined by the percentile $\alpha/2$ and $1 - \alpha/2$ of $\hat{y}_{n+\ell}$. The interval is $(L_l, L_u) = \left(\hat{F}(\alpha/2), \hat{F}(1 - \alpha/2) \right)$. We generate B bootstrap (for example, B = 1000) times series $Y_1^f, \dots, Y_2^f, \dots, Y_n^f$ by simulation from fitted SYMARMA(p, q) model. For each bootstrap series, $\hat{y}_{n+\ell}$ is calculated. From B bootstrap parameter estimates $\hat{y}_{n+\ell}^f$, we can form the empirical distribution $\hat{F}$.

6 An application

We consider the time series of excess return in the daily closing prices on Microsoft and the S&P 500 index. The S&P 500 was taken as the market price. This series consists of $n = 109$ observations from April 1, 2002 to September 5, 2002. The Microsoft's excess returns (Y_t) and the excess market return (X_t) were defined as

$$Y_t = r_t - r_{ft} \quad \text{and} \quad X_t = r_{mt} - r_{ft}$$

where r_t is the return¹ of Microsoft over a holding period t , r_{ft} is the risk-free return over the t th period evaluated by T-bill rates² and r_{mt} is the market return measured by S&P 500 index. A brief description of the complete dataset can be found in Ruppert (2004) (p. 239). The data and codes may be obtained from http://www.de.ufpe.br/~cysneiros/elliptical/time_series.html.

The maximum and minimum values for diary excess return of Microsoft are 11.10 and -9.47% , respectively. Figure 5a, b show the series of excess returns of Microsoft and the scatter plot for the two data sets, respectively. Figure 6a, b display the autocorrelation and partial autocorrelation functions plot of the series, respectively. The decay of the autocorrelation function and the plot in the Figure 5a suggests the stationarity of the series. The partial autocorrelation function plot suggest the autoregressive parameter of order 12. Initially, we fit the ARMA model, i.e. SYMARMA-normal model given by

$$Y_t = \beta X_t + \phi\{Y_{t-12} - \beta X_{t-12}\} + r_t, \quad t = 13, \dots, 109. \quad (9)$$

¹ The return is defined as $r_t = (p_t - p_{t-1})/p_{t-1}$ where p_t is the price of an asset at time t .

² The T-bill rates were divided by 100 to convert from a percentage and then by 253 to convert to a daily rate.

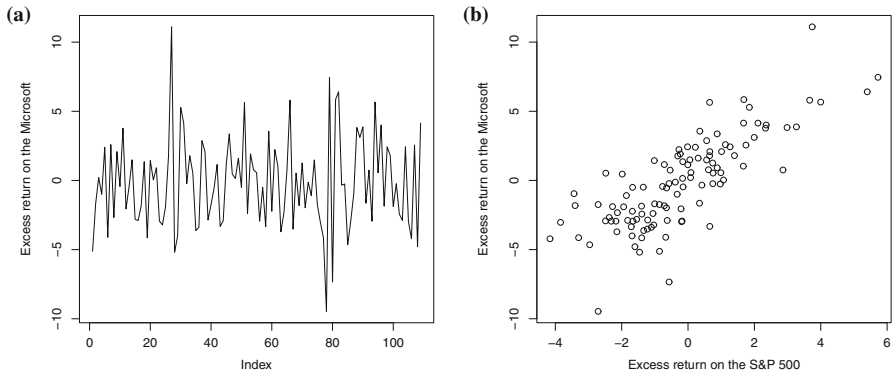


Fig. 5 Excess return on the Microsoft (a) and the relationship between the excess return on the Microsoft and on the S&P 500 index (b)

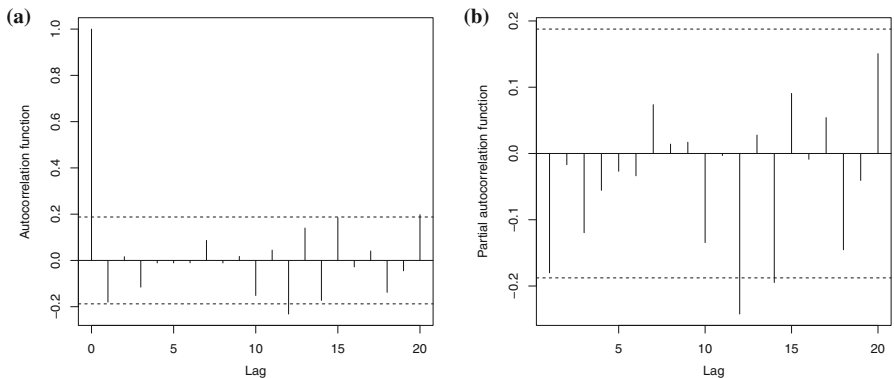


Fig. 6 Autocorrelation (a) and partial autocorrelation (b) functions of the excess return on the Microsoft

Chen and Liu (1993) developed a procedure to identify some types of the outliers in ARIMA models. Following this procedure, the time points $y_{27} = 11.10$ and $y_{80} = -7.34$ are considered additive outliers. The histogram, boxplot and normal probability plot of residuals of fitted SYMARMA-normal model displayed in Figs. 7 and 9a, respectively. From these figures, we can consider symmetry and heavy tails distribution for this data. In addition, the empirical skewness and kurtosis coefficients are -0.227 and 4.465 , respectively.

This way, we propose refit the model (9) under two heavy tails distributions: Student t and power exponential distributions. We use AIC and BIC criteria to select the extra parameters model, d.f. in Student t and κ in power exponential models. The RMSE, AIC and BIC values are evaluated (see Table 6). We fit the SYMARMA models under Student t distribution for some in the range $[3;15]$ and selected d.f. is equal to 5. Similarly, we fit SYMARMA models using the power exponential distribution, $PE(\kappa)$, for some κ values in the range $[0;1)$ (see Table 7). When $\kappa = 0$ and $=1$ we have the normal distribution and Laplace distribution, respectively. When κ parameter increase the kurtosis of the distribution increase. The selected fitted model has $\kappa = 0.3$.

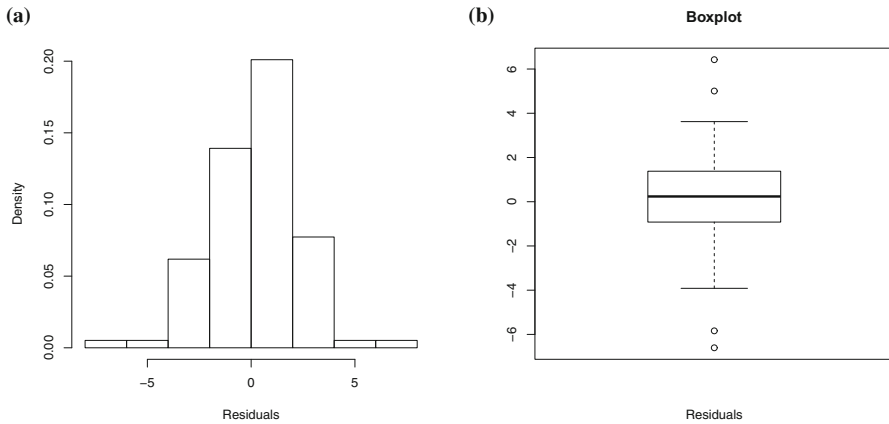


Fig. 7 Histogram (a) and boxplot of residuals (b) for the fitted SYMARMA-normal model

Table 6 Values AIC, BIC and RMSE for some fitted SYMARMA- t models

Models	AIC	BIC	RMSE
SYMARMA- t_3	413.91	419.06	3.43
SYMARMA- t_4	412.26	417.41	3.36
SYMARMA-t_5	411.68	416.83	3.34
SYMARMA- t_8	411.74	416.89	3.35
SYMARMA- t_{12}	412.47	417.63	3.39
SYMARMA- t_{15}	412.96	418.11	3.43

Table 7 Values AIC, BIC and RMSE for some fitted SYMARMA-EP models

Models	AIC	BIC	RMSE
SYMARMA-PE (0.1)	414.49	419.64	3.66
SYMARMA-PE (0.3)	412.78	417.92	3.62
SYMARMA-PE (0.5)	412.58	417.73	3.64
SYMARMA-PE (0.7)	413.32	418.47	3.68
SYMARMA-PE (0.9)	414.49	419.63	3.98

Estimates and asymptotic standard error for the parameters of the three fitted models and values for some comparison measures are presented in Table 8. Note that the SYMARMA- t_5 model show a lower value for both AIC, BIC and RMSE measures. In addition, we can note the $\hat{\phi}$, $\hat{\beta}$ and $\hat{\varphi}$ estimates of the SYMARMA- t_5 model have smaller asymptotic estimated standard errors than SYMARMA-normal model. In economic area, β can be interpreted, marginally, as systematic risk, similar to linear regression model (see, [Ruppert 2004](#), Chapter 7 and [Paula and Cysneiros 2009](#)). In this example, the value $\hat{\beta}$ indicate an *agressive* capital asset.

Residuals correlation and partial autocorrelations function of the fitted SYMARMA- t_5 and the normal probability plot with envelopes indicate a good adequacy (see Figs.

Table 8 Estimates and standard errors (in parenthesis) for the parameters of three fitted models and some comparison measures

Model	Parameter	Estimates	AIC	BIC	RMSE
SYMARMA-normal	β	1.327(0.105)	416.20	421.35	3.73
	ϕ	-0.132(0.100)			
	φ	4.103(0.589)			
SYMARMA- t_5	β	1.273(0.097)	411.68	416.83	3.34
	ϕ	-0.162(0.093)			
	φ	2.628(0.477)			
SYMARMA-PE (0.3)	β	1.299(0.100)	412.78	417.92	3.62
	ϕ	-0.134(0.096)			
	φ	2.310(0.378)			

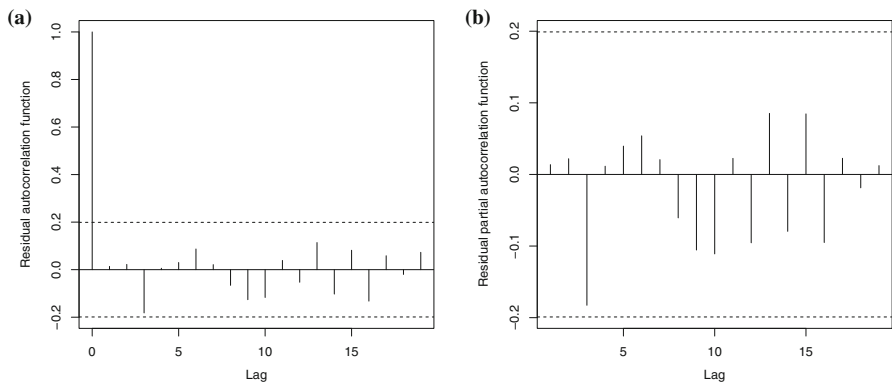


Fig. 8 Residual autocorrelation (a) and partial autocorrelation (b) functions of SYMARMA- t_5 model

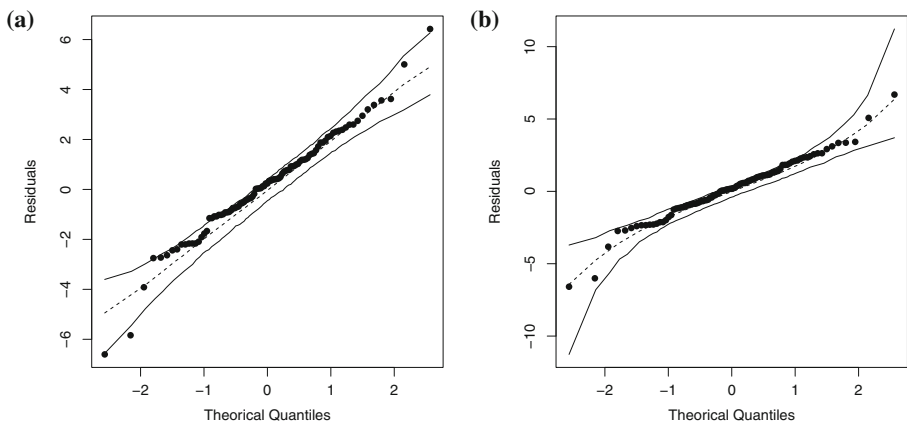


Fig. 9 Normal probability plot with envelopes for fitted SYMARMA-normal (a) and SYMARMA- t_5 model (b)

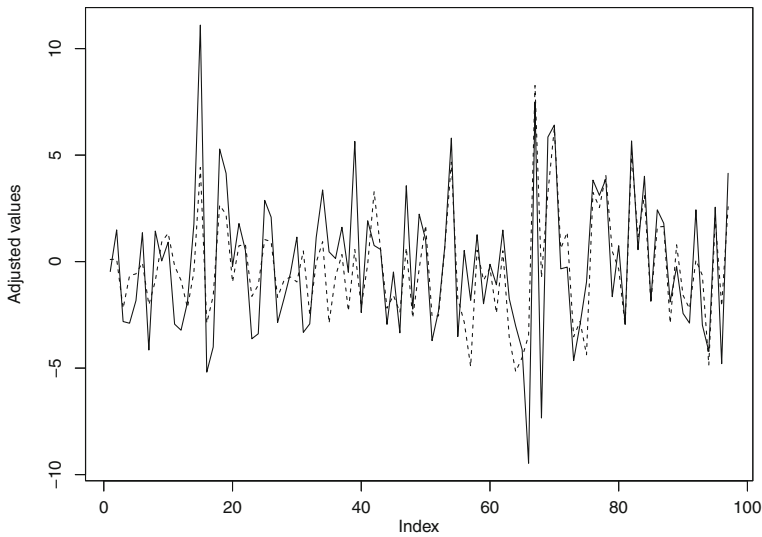


Fig. 10 Original series (*solid line*) and estimated curve (*dashed line*) through an fitted SYMARMA- t_5 model

Table 9 Forecasts 12 steps ahead, limits of 95 % bootstrap confidence interval and RMSE from fitted SYMARMA model

ℓ	$y_{n+\ell}$	SYMARMA-normal				SYMARMA- t_5			
		$\hat{y}_{n+\ell}$	L_l	L_u	RMSE	$\hat{y}_{n+\ell}$	L_l	L_u	RMSE
1	1.84	1.24	1.09	1.61	0.33	1.15	1.09	1.53	0.38
2	2.26	0.97	0.84	1.13	0.47	0.92	0.83	1.05	0.50
3	-2.46	-0.18	-0.26	0.20	0.66	-0.19	-0.23	0.15	0.67
4	-2.94	-3.14	-3.93	-2.78	0.57	-2.97	-3.67	-2.77	0.58
5	1.61	0.50	0.30	0.56	0.60	0.52	0.30	0.55	0.60
6	-1.31	-2.38	-2.87	-2.10	0.64	-2.25	-2.69	-2.07	0.62
7	0.97	-0.95	-1.13	-0.14	0.95	-1.00	-1.09	-0.18	0.96
8	-1.18	-3.64	-4.88	-3.21	1.16	-3.40	-4.62	-3.23	1.11
9	0.55	0.15	0.05	0.57	1.17	0.13	0.09	0.51	1.08
10	-4.70	-1.88	-2.16	-1.59	1.08	-1.82	-2.02	-1.57	1.04
11	0.93	-1.95	-2.98	-1.68	1.39	-1.76	-2.82	-1.65	1.33
12	1.88	3.04	2.68	3.98	1.34	2.83	2.69	3.76	1.28

8, 9b). The test *Ljung-Box* (see [Ljung and Box 1978](#)) was used to check the randomness of residuals in the fitted SYMARMA- t_5 models. The test indicates that the residuals are not autocorrelated at 5 % significance (p value equal to 0.891). Figure 10 display the fitted SYMARMA- t_5 series against the original series indicating a good agreement.

Table 9 presents the twelve forecasts steps ahead, limits of 95 % bootstrap confidence interval and RMSE from fitted SYMARMA model. Based on RMSE, The fitted SYMARMA-normal has better performance in short time ahead while fitted SYMARMA- t_5 for a long time ahead.

7 Concluding remarks

In this paper, a new time series model is introduced named SYMARMA which is a dynamic model for random variables belonging to the class of symmetric distributions. We discuss methods for parameter estimation, hypothesis testing and forecasting. The simulation studies showed a better performance for the SYMARMA(1,0) model under conditional t_4 distribution in terms of qualitative robustness of the estimation procedure. In the application, fitted SYMARMA(1,0) model under conditional t_5 has the better performance in the additive outliers.

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Appendix 1: Proof of Theorem 1

Proof Let $\Phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p$ the autoregressive polynomial, $\Theta(B) = 1 + \theta_1 B + \dots + \theta_q B^q$ the moving averages polynomial and $B^k y_t = y_{t-k}$ the lag operator and $\Psi(B) = \sum_{i=0}^{\infty} \psi_i B^i = \Theta(B)\Phi(B)^{-1}$, $\psi_0 = 1$ and assuming that $\Phi(B)$ is invertible. The SYMARMA model can be rewritten as

$$\Phi(B)(y_t - \mathbf{x}_t^\top \boldsymbol{\beta}) = \Theta(B)r_t$$

and, since $\Phi(B)$ is invertible,

$$y_t = \mathbf{x}_t^\top \boldsymbol{\beta} + \Psi(B)r_t,$$

Therefore, assuming that $\Phi(B)$ is invertible, the marginal mean of y_t of the SYMARMA model is given by

$$E(y_t) = \mathbf{x}_t^\top \boldsymbol{\beta}.$$

□

Appendix 2: Proof of Theorem 2

Proof Let $Y_t = \mu_t + r_t$ where r_t 's are uncorrelated residuals with mean zero. We have

$$\begin{aligned} \text{Var}(r_t) &= E(r_t^2) = E(E(r_t^2 | \mathcal{F}_{t-1})) = E(\text{Var}(r_t | \mathcal{F}_{t-1})) = E(\text{Var}(Y_t - \mu_t | \mathcal{F}_{t-1})) \\ &= E(\text{Var}(y_t | \mathcal{F}_{t-1})) = E(\xi \varphi) = \xi \varphi. \end{aligned}$$

Note that μ_t given in (2) is $\mathcal{F}_{t-1}$ -measurable. Therefore, the marginal variance of Y_t , $\text{Var}(Y_t)$, is given by

$$\begin{aligned}\text{Var}(Y_t) &= \text{Var}(\mathbf{x}_t^\top \boldsymbol{\beta} + \Psi(B)r_t) = \text{Var}(\Psi(B)r_t) = \mathbb{E}[(\Psi(B)r_t)^2] \\ &= \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \psi_i \psi_j \mathbb{E}(r_{t-i} r_{t-j}) = \sum_{i=0}^{\infty} \psi_i^2 \mathbb{E}(r_{t-i}^2) = \sum_{i=0}^{\infty} \psi_i^2 \text{Var}(r_{t-i}) \\ &= \xi \varphi \sum_{i=0}^{\infty} \psi_i^2.\end{aligned}$$

□

Appendix 3: Proof of Theorem 3

Proof By Theorem 1 and 2 we have

$$\begin{aligned}\mathbb{E}(Y_t) &= \mathbf{x}_t^\top \boldsymbol{\beta} \quad \text{e} \quad \text{Var}(Y_t) = \xi \varphi \sum_{i=0}^{\infty} \psi_i^2. \quad \text{And} \\ \text{Cov}(Y_t, Y_{t-k}) &= \text{Cov}(\mathbf{x}_t^\top \boldsymbol{\beta} + \sum_{i=0}^{\infty} \psi_i r_{t-i}, \mathbf{x}_{t-k}^\top \boldsymbol{\beta} + \sum_{i=0}^{\infty} \psi_i r_{t-k-i}) \\ &= \text{Cov}(\psi_0 r_t + \psi_1 r_{t-1} + \cdots, \psi_0 r_{t-k} + \psi_1 r_{t-k-1} + \cdots) \\ &= \text{Var}(r_t) \sum_{i=0}^{\infty} \psi_i \psi_{i+k} = \xi \varphi \sum_{i=0}^{\infty} \psi_i \psi_{i+k}\end{aligned}$$

where $\Psi_0 = 1$.

$$\begin{aligned}\text{Corr}(Y_t, Y_{t-k}) &= \frac{\text{Cov}(Y_t, Y_{t-k})}{\sqrt{\text{Cov}(Y_t, Y_t) \text{Cov}(Y_{t-k}, Y_{t-k})}} = \frac{\xi \varphi \sum_{i=0}^{\infty} \psi_i \psi_{i+k}}{\xi \varphi \sum_{i=0}^{\infty} \psi_i^2} \\ &= \frac{\sum_{i=0}^{\infty} \psi_i \psi_{i+k}}{\sum_{i=0}^{\infty} \psi_i^2}.\end{aligned}$$

□

Expected conditional Fisher information matrix

The elements of the expected conditional Fisher information matrix, $\mathbf{K}$, are obtained from expression

$$\mathbf{K}_{\omega_r \omega_s} = -\mathbb{E} \left[\frac{\partial^2 \ell(\boldsymbol{\delta}, \varphi)}{\partial \omega_r \partial \omega_s} \middle| \mathcal{F}_{t-1} \right] = \mathbb{E} \left[\frac{\partial \ell(\boldsymbol{\delta}, \varphi)}{\partial \omega_r} \frac{\partial \ell(\boldsymbol{\delta}, \varphi)}{\partial \omega_s} \middle| \mathcal{F}_{t-1} \right],$$

where ω_r and ω_s are model parameters and ℓ is the logarithm of the conditional likelihood function.

Under suitable regularity conditions

$$\begin{aligned}
 \mathbb{E} \left(\frac{\partial \ell_t(\boldsymbol{\delta}, \varphi)}{\partial \mu_t} \mid \mathcal{F}_{t-1} \right) &= \mathbb{E} \left(\frac{\partial \log f(y_t | \mathcal{F}_{t-1})}{\partial \mu_t} \right) \\
 &= \int_{-\infty}^{\infty} \frac{\partial \log f(y_t | \mathcal{F}_{t-1})}{\partial \mu_t} f(y_t | \mathcal{F}_{t-1}) d\mu_t \\
 &= \int_{-\infty}^{\infty} \left(\frac{1}{f(y_t | \mathcal{F}_{t-1})} \frac{\partial f(y_t | \mathcal{F}_{t-1})}{\partial \mu_t} \right) f(y_t | \mathcal{F}_{t-1}) d\mu_t \\
 &= \int_{-\infty}^{\infty} \frac{\partial f(y_t | \mathcal{F}_{t-1})}{\partial \mu_t} d\mu_t = \frac{\partial}{\partial \mu_t} \int_{-\infty}^{\infty} f(y_t | \mathcal{F}_{t-1}) d\mu_t = 0.
 \end{aligned} \tag{10}$$

From some algebraic manipulations, we also obtain that

$$\frac{\partial \ell_t(\boldsymbol{\delta}, \varphi)}{\partial \mu_t} = -\frac{2}{\sqrt{\varphi}} W_g(u_t) z_t,$$

with $z_t = \sqrt{u_t} = (y_t - \mu_t) / \sqrt{\varphi}$. Therefore, using the results in (10), we have

$$\mathbb{E} (W_g(u_t) z_t | \mathcal{F}_{t-1}) = 0. \tag{11}$$

Furthermore, the expressions

$$\frac{\partial \mu_t}{\partial \beta_l} = x_{tl} - \sum_{i=1}^p \phi_i x_{(t-i)l}, \quad \frac{\partial \mu_t}{\partial \phi_i} = y_{t-i} - \mathbf{x}_{t-i}^\top \boldsymbol{\beta} \quad \text{and} \quad \frac{\partial \mu_t}{\partial \theta_j} = y_{t-j} - \mu_{t-j} \tag{12}$$

are measurable with respect to $\mathcal{F}_{t-1}$.

The $\mathbf{K}_{\delta\delta}$ matrix elements

$$\begin{aligned}
 &\mathbb{E} \left(\frac{\partial \ell_t(\boldsymbol{\delta}, \varphi)}{\partial \delta_i} \frac{\partial \ell_t(\boldsymbol{\delta}, \varphi)}{\partial \delta_j} \mid \mathcal{F}_{t-1} \right) \\
 &= \mathbb{E} \left[\left(\frac{-2W_g(u_t)}{\sqrt{\varphi}} \frac{\partial \mu_t}{\partial \delta_i} z_t \right) \left(\frac{-2W_g(u_t)}{\sqrt{\varphi}} \frac{\partial \mu_t}{\partial \delta_j} z_t \right) \mid \mathcal{F}_{t-1} \right] \\
 &= \frac{4}{\varphi} \mathbb{E} \left[W_g^2(u_t) z_t^2 \frac{\partial \mu_t}{\partial \delta_i} \frac{\partial \mu_t}{\partial \delta_j} \mid \mathcal{F}_{t-1} \right] \\
 &= \frac{4}{\varphi} \mathbb{E} \left[W_g^2(u_t) z_t^2 \mid \mathcal{F}_{t-1} \right] \frac{\partial \mu_t}{\partial \delta_i} \frac{\partial \mu_t}{\partial \delta_j} \\
 &= \frac{4}{\varphi} d_g \frac{\partial \mu_t}{\partial \delta_i} \frac{\partial \mu_t}{\partial \delta_j},
 \end{aligned}$$

with $d_g = E \left[W_g^2(u_t) z_t^2 | \mathcal{F}_{t-1} \right]$. So, $d_g = E \left[W_g^2(U^2) U^2 | \mathcal{F}_{t-1} \right]$ with $U \sim S(0, 1, g)$.

From the results presented in (12) one can easily find the expressions for the elements of $\mathbf{K}_{\delta\delta}$.

The $\mathbf{K}_{\varphi\varphi}$ matrix elements

$$\begin{aligned} E \left(\frac{\partial \ell_t(\boldsymbol{\delta}, \varphi)}{\partial \varphi} \frac{\partial \ell_t(\boldsymbol{\delta}, \varphi)}{\partial \varphi} | \mathcal{F}_{t-1} \right) &= E \left[\left(-\frac{1}{2\varphi} - \frac{W_g(u_t)}{\varphi} u_t \right) \left(-\frac{1}{2\varphi} - \frac{W_g(u_t)}{\varphi} u_t \right) | \mathcal{F}_{t-1} \right] \\ &= E \left[\frac{1}{4\varphi^2} + \frac{W_g(u_t) u_t}{\varphi^2} + \frac{W_g^2(u_t) u_t^2}{\varphi^2} | \mathcal{F}_{t-1} \right] \\ &= \frac{1}{4\varphi^2} + \frac{1}{\varphi^2} E \left[W_g(u_t) u_t | \mathcal{F}_{t-1} \right] + \frac{1}{\varphi^2} E \left[W_g^2(u_t) u_t^2 | \mathcal{F}_{t-1} \right] \\ &= \frac{1}{4\varphi^2} + \frac{1}{\varphi^2} \left(-\frac{1}{2} \right) + \frac{1}{\varphi^2} f_g \\ &= \frac{1}{\varphi^2} f_g - \frac{1}{4\varphi^2} = \frac{1}{4\varphi^2} (4f_g - 1), \end{aligned}$$

with $f_g = E \left[W_g^2(u_t) u_t^2 | \mathcal{F}_{t-1} \right]$. So, $f_g = E \left[W_g^2(U^2) U^4 | \mathcal{F}_{t-1} \right]$ with $U \sim S(0, 1, g)$.

From Fang et al. (1990) (p. 94). we have $E \left[W_g(u_t) u_t | \mathcal{F}_{t-1} \right] = -1/2$. Therefore,

$$\mathbf{K}_{\varphi\varphi} = \sum_{t=m+1}^n \frac{1}{4\varphi^2} (4f_g - 1) = \frac{(n-m)}{4\varphi^2} (4f_g - 1).$$

$\mathbf{K}_{\delta\varphi}$ matrix

$$\begin{aligned} E \left(\frac{\partial \ell_t(\boldsymbol{\delta}, \varphi)}{\partial \delta_i} \frac{\partial \ell_t(\boldsymbol{\delta}, \varphi)}{\partial \varphi} | \mathcal{F}_{t-1} \right) &= E \left[\left(\frac{-2W_g(u_t)}{\sqrt{\varphi}} \frac{\partial \mu_t}{\partial \delta_i} z_t \right) \left(-\frac{1}{2\varphi} - \frac{W_g(u_t)}{\varphi} u_t \right) | \mathcal{F}_{t-1} \right] \\ &= E \left[\frac{W_g(u_t)}{\varphi \sqrt{\varphi}} z_t | \mathcal{F}_{t-1} \right] \frac{\partial \mu_t}{\partial \delta_i} + E \left[\frac{2W_g^2(u_t)}{\varphi \sqrt{\varphi}} z_t u_t | \mathcal{F}_{t-1} \right] \frac{\partial \mu_t}{\partial \delta_i} \\ &= \frac{1}{\varphi \sqrt{\varphi}} \left\{ E \left[W_g(u_t) z_t | \mathcal{F}_{t-1} \right] + 2E \left[W_g^2(u_t) z_t u_t | \mathcal{F}_{t-1} \right] \right\} \frac{\partial \mu_t}{\partial \delta_i} \\ &= 0. \end{aligned}$$

From Fang et al. (1990) (p. 94) we have $E \left[W_g^2(u_t) z_t u_t | \mathcal{F}_{t-1} \right] = 0$ and in addition, $E \left[W_g(u_t) z_t | \mathcal{F}_{t-1} \right] = 0$ because (11).

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